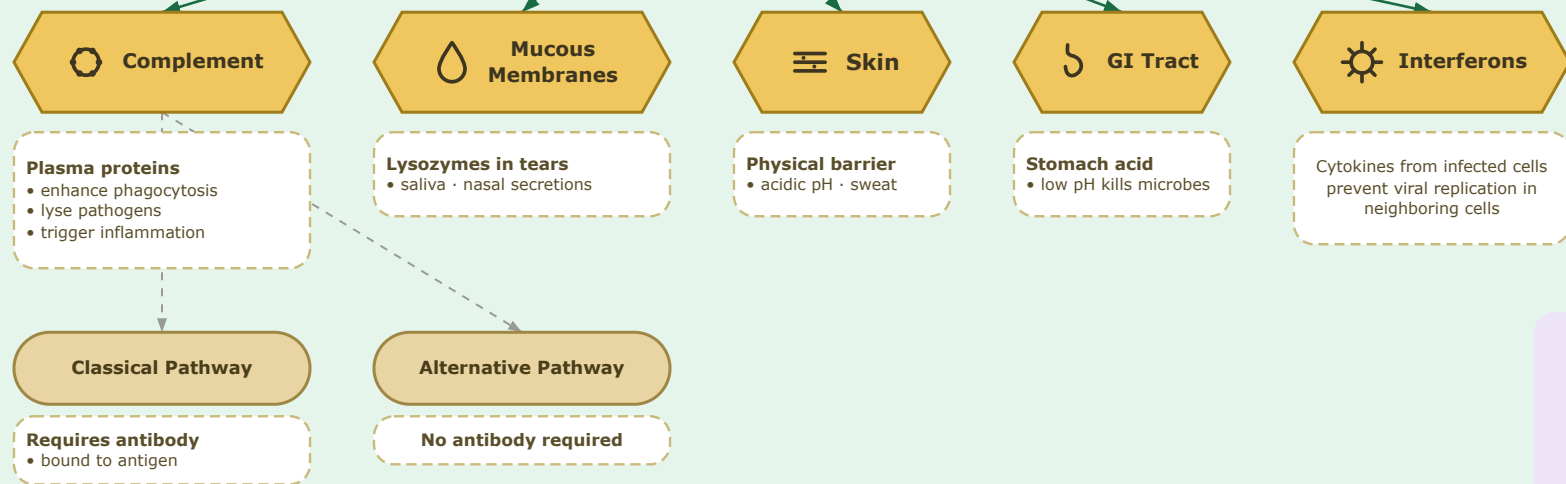


IMMUNE SYSTEM

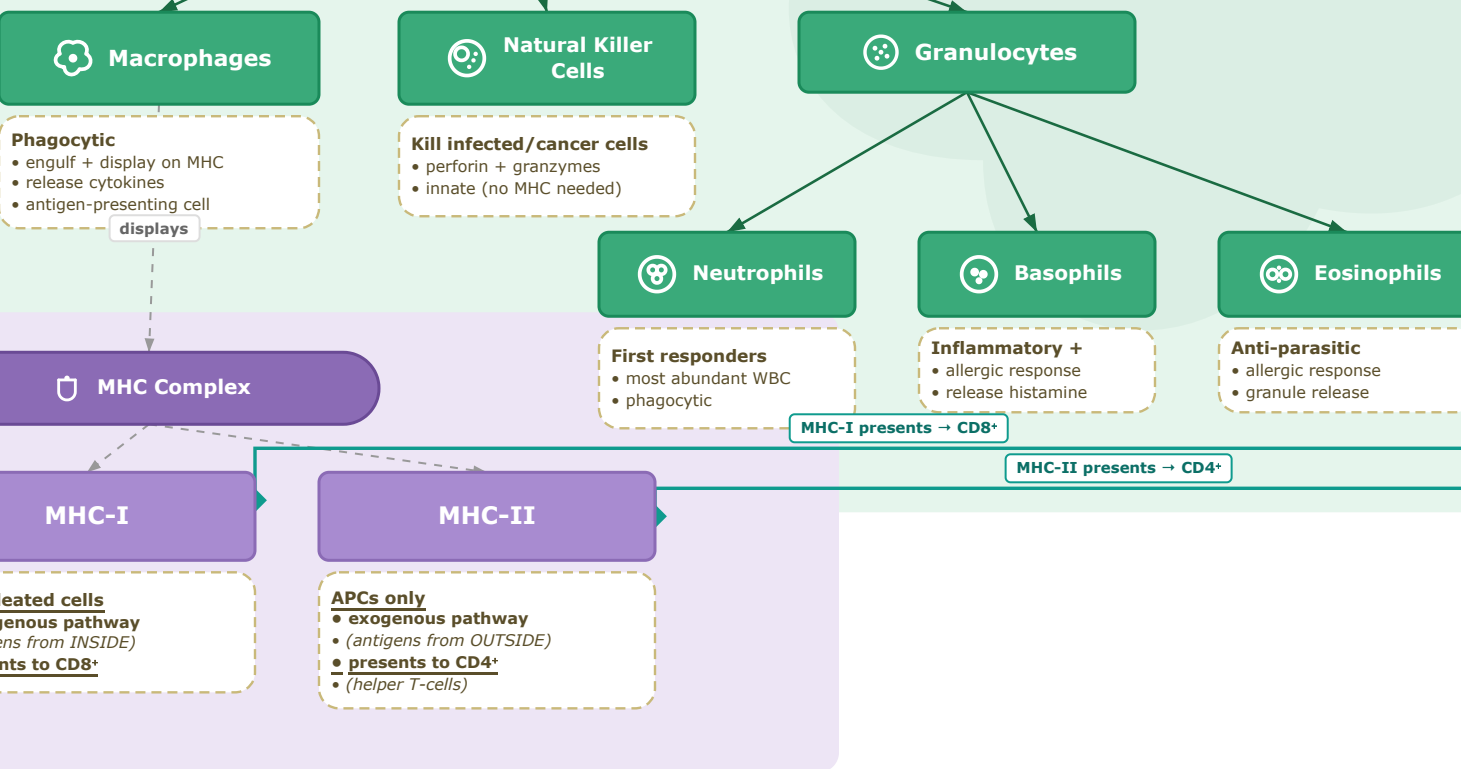
INNATE IMMUNE SYSTEM

Always active · non-specific · no memory

NONCELLULAR DEFENSES



CELLS OF INNATE SYSTEM

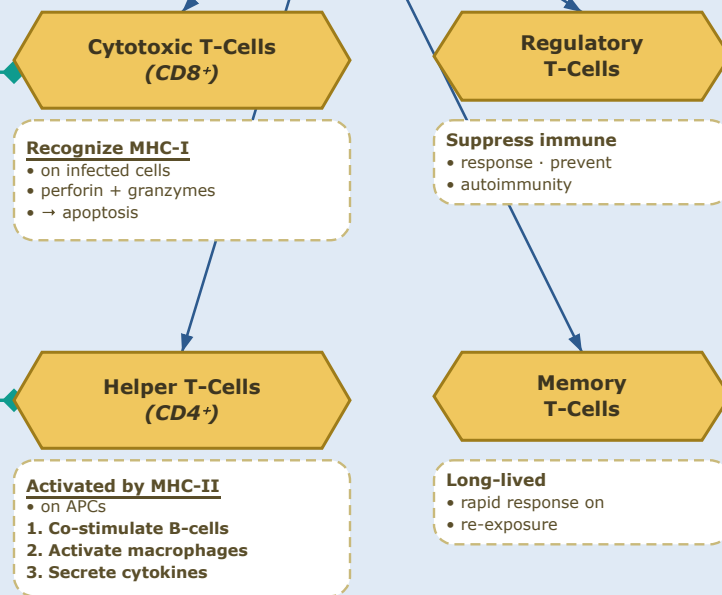


ADAPTIVE IMMUNE SYSTEM

Specific · slower · has memory

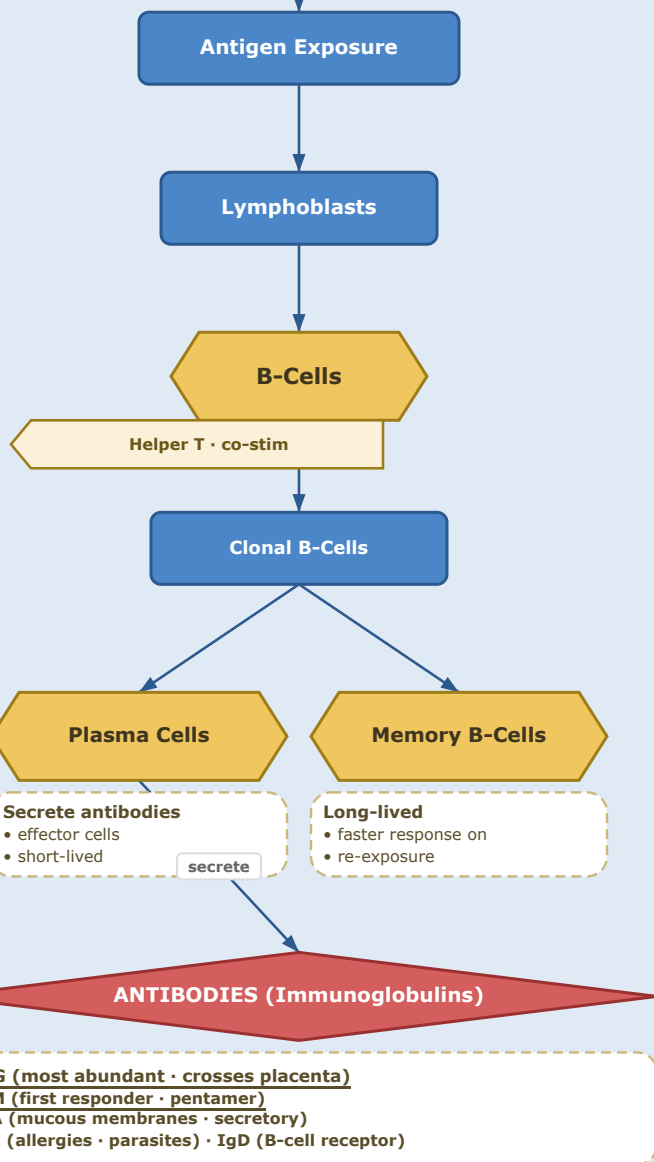
CELL-MEDIATED

T-cell mediated



HUMORAL IMMUNITY

Antibody-mediated (B-cells)



KEY: Innate = fast · non-specific · no memory Adaptive = slow · specific · HAS memory
Humoral = B-cells + antibodies (extracellular) Cell-mediated = T-cells (intracellular pathogens)

MHC-I + CD8⁺ = cytotoxic (kill infected cells) MHC-II + CD4⁺ = helper (coordinate response)

LEGEND

○ system □ cell type ◇ key concept ◊ purple = MHC — note/cross-ref
→ green = innate → blue = adaptive ↔ teal = MHC↔T-cell cooperation > gold chip = cross-reference underline = high-yield